

The Accuracy–Transparency Trade-Off in Responsible AI

Umut Türklay · 49848742 · Governing Intelligent Systems for Sustainability

1 Motivation

High-stakes automated decisions — credit, hiring, risk scoring — increasingly rely on machine learning. Regulations such as the EU AI Act and the GDPR right to explanation demand transparency, which sits in tension with more accurate but opaque black-box models. This project measures how real, how large, and how fair that trade-off actually is on real data.

2 Dataset

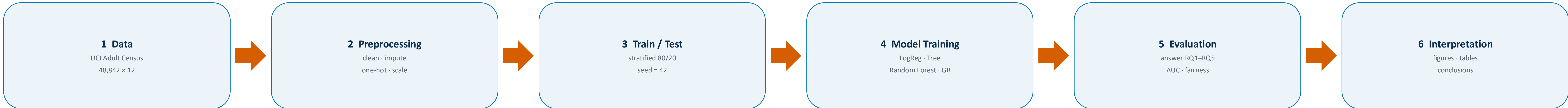
UCI Adult Census Income

- 48,842 records, 12 features (after cleaning)
- Target: income > 50K (positive rate 23.9%)
- Source: OpenML / Kaggle (public)
- Dropped fnlwgt (sampling weight) and duplicate education column
- Sensitive attributes available: sex, race

3 Research Questions

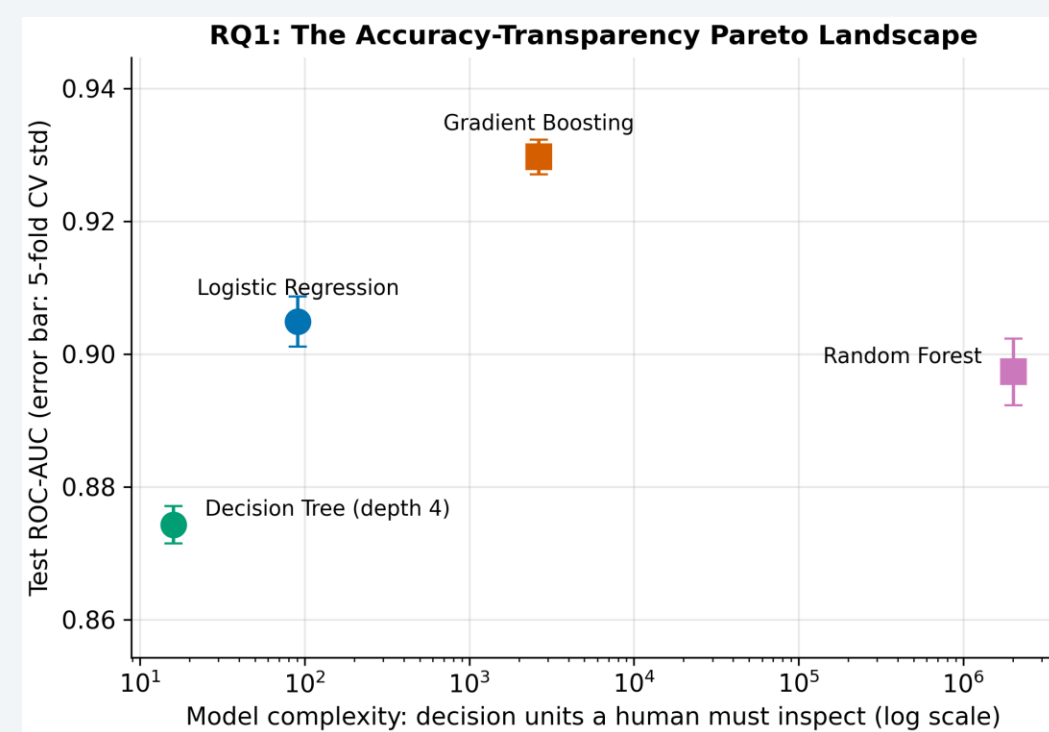
RQ1 How large is the accuracy gap between interpretable and black-box models?
RQ2 Where do added complexity and lost transparency stop paying off?
RQ3 Do the models rely on the same features?
RQ4 Is the black-box advantage uniform across demographic subgroups?
RQ5 How faithfully can a compact surrogate mimic the black-box?

4 Methodology Pipeline



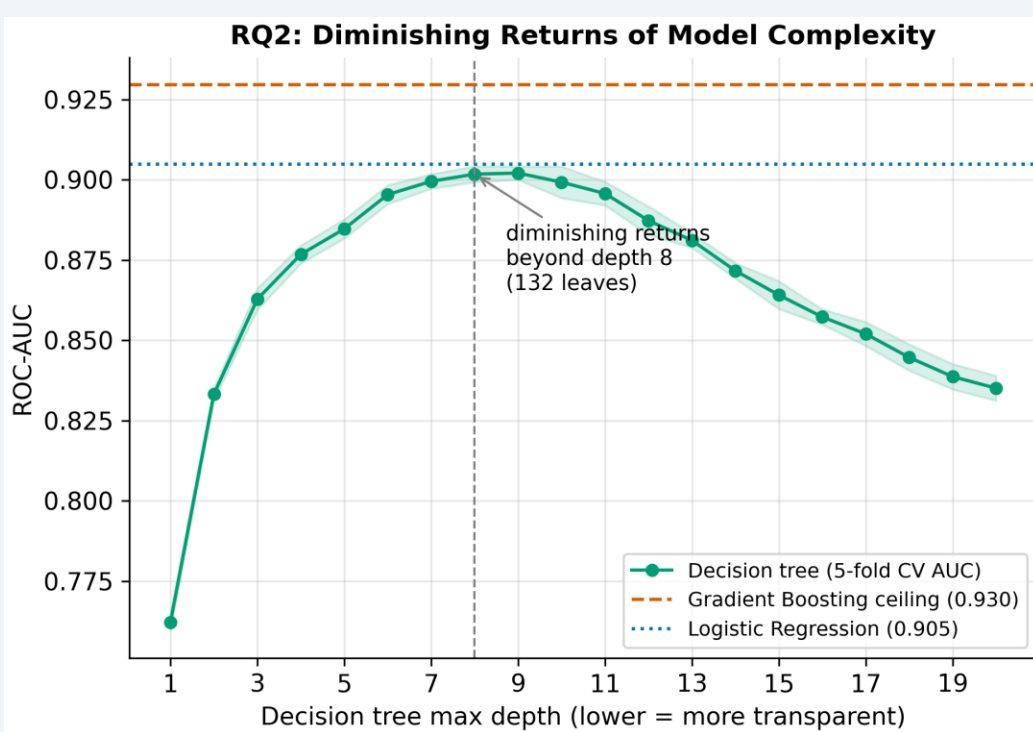
5 Key Results

RQ1 · Accuracy vs Transparency



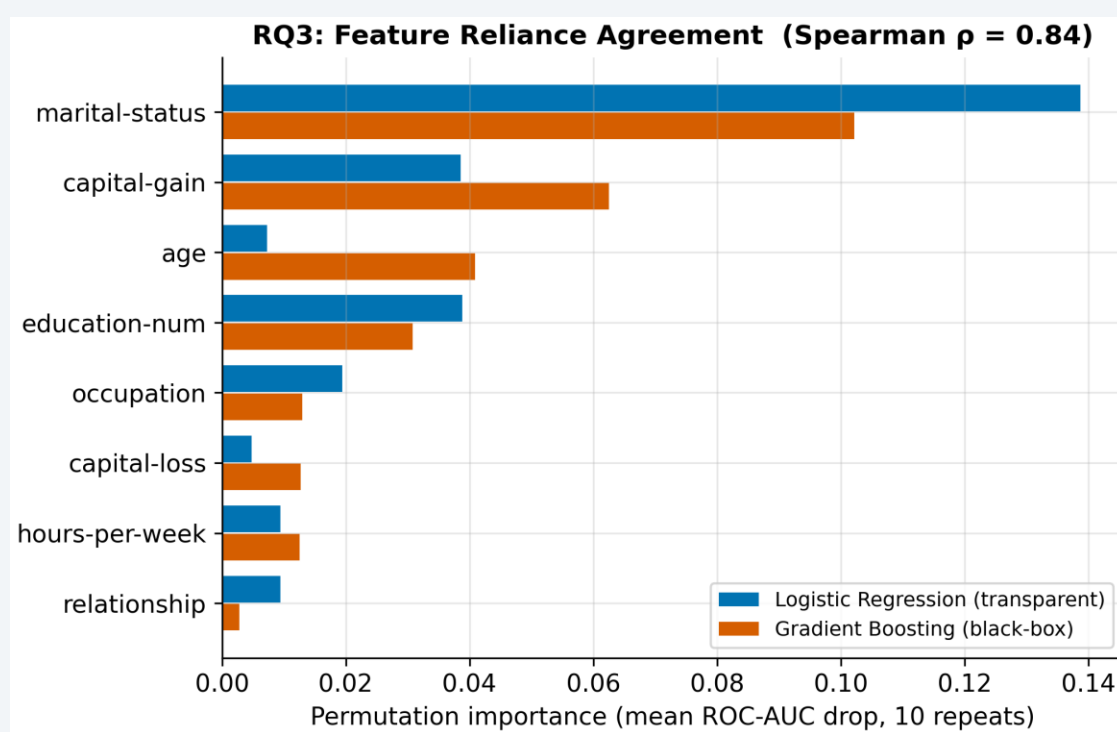
Black-box edge is real but small: GB AUC 0.930 vs LogReg 0.905 (+2.5 pts). Random Forest uses ~2M leaves yet trails — complexity ≠ accuracy.

RQ2 · Diminishing Returns



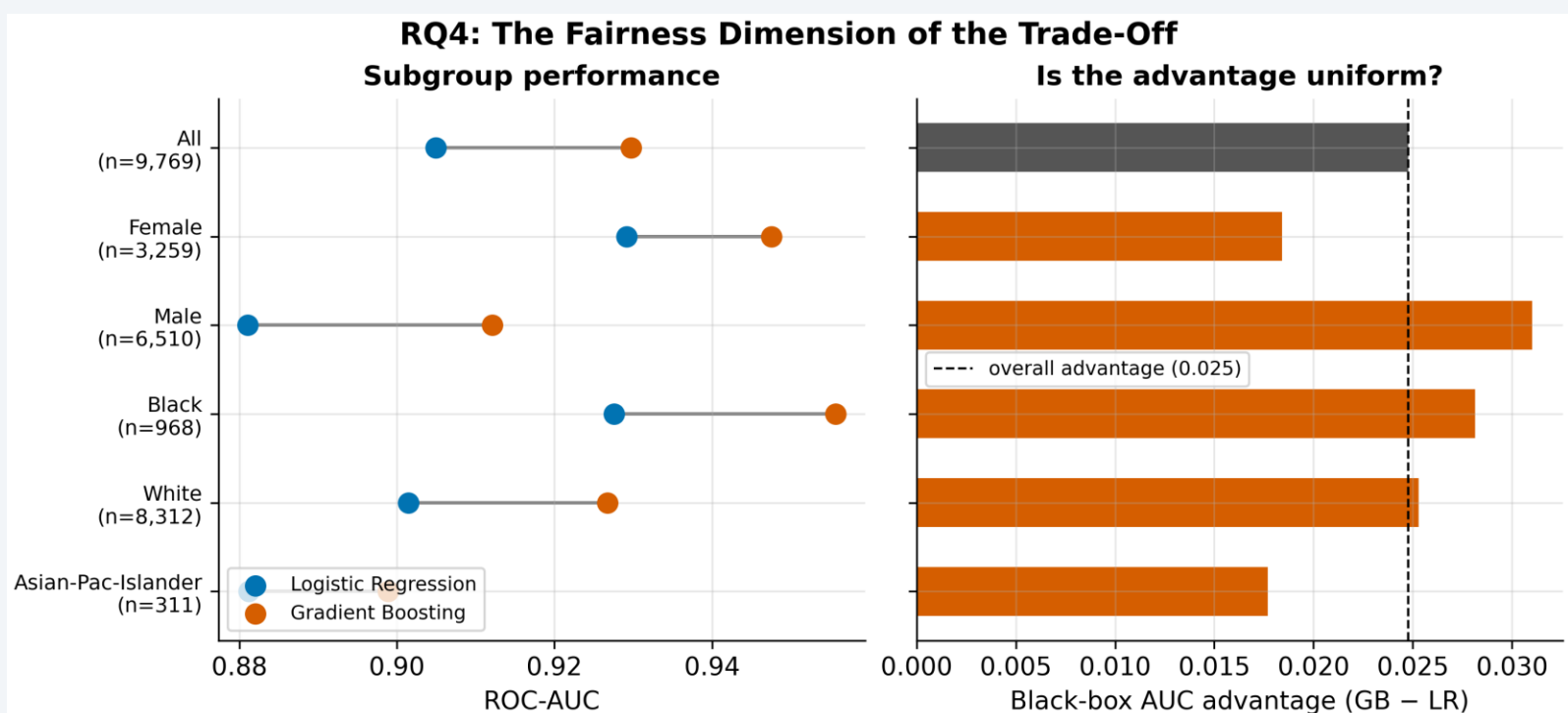
A depth-8 tree (132 leaves) already matches LogReg; beyond it accuracy falls while transparency keeps degrading.

RQ3 · Feature Agreement



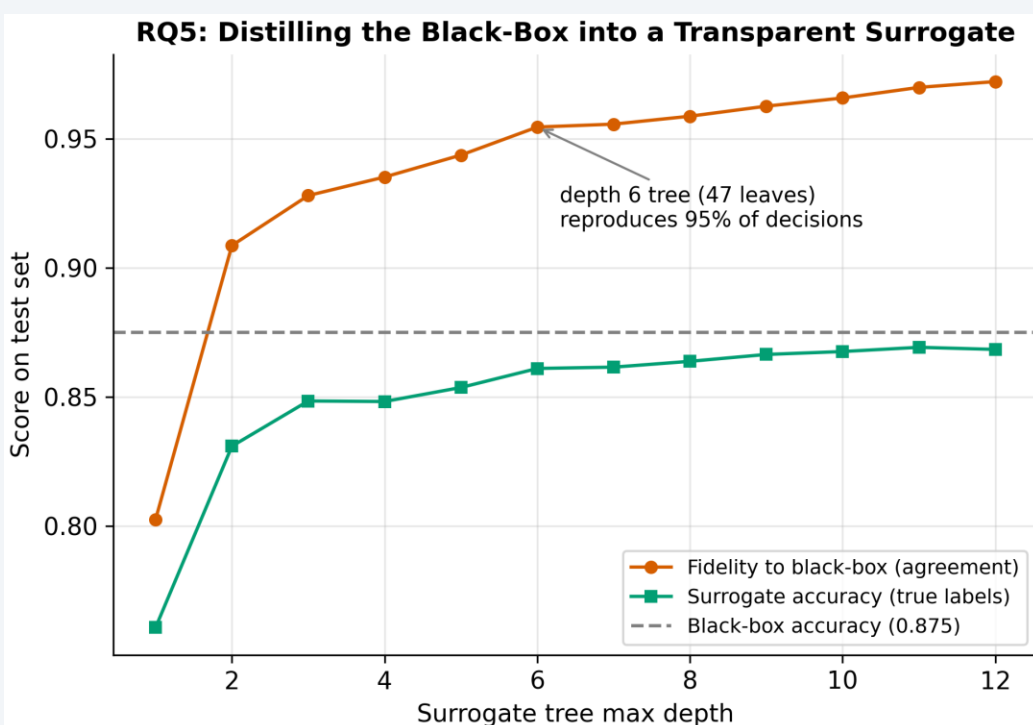
Both model types rely on largely the same features (Spearman $\rho = 0.84$); GB mainly differs by exploiting the non-linear age effect.

RQ4 · Fairness Dimension



The advantage is NOT uniform: +0.031 AUC for men vs +0.018 for women. Choosing transparency costs subgroups differently.

RQ5 · Surrogate Distillation



A depth-6 surrogate (47 leaves) reproduces 95.4% of black-box decisions at 0.861 accuracy vs the teacher's 0.875 — an auditable middle ground.

6 Contributions

- A measurable transparency axis (decision units)
- Quantified trade-off size on real data (~2.5 AUC pts)
- Evidence that complexity alone ≠ accuracy
- A fairness-aware view of the trade-off
- Distillation as a practical, auditable remedy

Discussion

The black-box gain is modest and, under default settings, Random Forest even trails logistic regression — accuracy is not a simple function of complexity. Because the advantage is unequal across groups, switching to a black-box is also a governance decision, not only a technical one. Limitations: one dataset, default hyperparameters, a single transparency proxy.

7 Conclusion

The accuracy–transparency trade-off is real but small (~2.5 AUC points), it is not distributed equally across demographic groups, and it can be largely closed by distilling the black-box into a compact, auditable surrogate.